

Mark Scheme

Q1.

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	$\frac{523 + 52 + 28}{184} = \frac{132}{184} = 0.717$	B1	This mark is given for a correct value for the probability for the cloud cover
(b)(i)	$P(X \geq 6) = 1 - P(X \leq 5)$	M1	This mark is given for using $1 - P(X \leq 5)$ with $B(8, 0.76)$
	$= 1 - 0.2967$ $= 0.703$	A1	This mark is given for finding as correct value for the probability
(b)(ii)	$184 \times P(X = 7)$ $= 184 \times 0.2811$	M1	This mark is given for using $184 \times P(X = 7)$ with $B(8, 0.76)$
	$= 51.7$	A1	This mark is given for finding as correct value for the probability
(c)	The answer to part (b)(i) of 0.703 is similar to 0.7127 in part (a) The answer to part (b)(ii) of 51.7 is very close to 52 found in the data set	B1	This mark is given for a correct evaluation of the outcomes from part (b) to determine the appropriateness of Magali's model
(d)	$\frac{5 + 9 + 9}{28} = \frac{23}{28} = 0.821$	B1	This mark is given for a correct value for the probability for the cloud cover
(e)	The answer to part (d) of 0.821 is greater than that in part (a) of 0.717 This shows that there is a higher chance of having high cloud cover if the previous day had high cloud cover	B1	This mark is given for a correct comparison for the answer to part (d) with the data set
	Thus independence does not hold so a binomial model might not be suitable	B1	This mark is given for a correct conclusion stated
(Total 9 marks)			

Q2.

	Scheme	Marks	AO												
(a)	Disadvantage: e.g. Not random; cannot use (reliably) for inferences	B1	1.1b												
(b)	[Sight or correct use of] $X \sim B(36, 0.08)$	M1	3.3												
(i)	$P(X=4) = 0.167387\dots$ awrt <u>0.167</u>	A1	1.1b												
(ii)	$[P(X \leq 7) = 1 - P(X \geq 6)]$ 0.022233... awrt <u>0.0222</u>	A1	1.1b												
(c)	$P(\text{In dance club and dance tango}) = 0.4 \times 0.08 = \underline{0.032}$ or $\frac{4}{125}$ or <u>3.2%</u>	B1	1.1b												
(d)	[Let T = those who can dance the Tango. Sight or use of] $T \sim B(50, "0.032")$ $[P(T < 3) = P(T \leq 2)]$ 0.7850815... awrt <u>0.785</u>	M1 A1	3.3 1.1b												
		(2)													
		(7 marks)													
	Notes														
(a)	B1 for a suitable disadvantage:														
	<table border="1"> <thead> <tr> <th>Allow (B1)</th> <th>Do NOT allow (B0)</th> </tr> </thead> <tbody> <tr> <td>Not random <u>or</u> less random (o.e.)</td> <td>Not representative</td> </tr> <tr> <td>Cannot use (reliably) for inferences</td> <td>Less accurate</td> </tr> <tr> <td>(More likely to be) biased</td> <td>Any comment based on time or cost</td> </tr> <tr> <td></td> <td>Any mention of skew</td> </tr> <tr> <td></td> <td>Any mention of non-response</td> </tr> </tbody> </table>	Allow (B1)	Do NOT allow (B0)	Not random <u>or</u> less random (o.e.)	Not representative	Cannot use (reliably) for inferences	Less accurate	(More likely to be) biased	Any comment based on time or cost		Any mention of skew		Any mention of non-response		
Allow (B1)	Do NOT allow (B0)														
Not random <u>or</u> less random (o.e.)	Not representative														
Cannot use (reliably) for inferences	Less accurate														
(More likely to be) biased	Any comment based on time or cost														
	Any mention of skew														
	Any mention of non-response														
(b)	M1 for sight of $B(36, 0.08)$ Allow in words: <u>binomial</u> with $n = 36$ and $p = 0.08$ may be implied by one correct answer to 2sf <u>or</u> sight of $P(X \leq 6) = 0.97776\dots$ i.e. awrt 0.98 Allow for $36C4 \times 0.08^4 \times 0.92^{32}$ as this is "correct use"														
(i)	1 st A1 for awrt 0.167 NB An answer of just awrt 0.167 scores M1($\Rightarrow$) 1 st A1														
(ii)	2 nd A1 for awrt 0.0222														
(c)	B1 for 0.032 o.e. (Can allow for sight of 0.4×0.08)														
(d)	M1 for sight of $B(50, "0.032")$ ft their answer to (c) provided it is a probability $\neq 0.08$ may be implied by correct answer <u>or</u> sight of $[P(T \leq 3)] = 0.924348\dots$ i.e. awrt 0.924 or $P(T \leq 2)$ as part of $1 - P(T \leq 2)$ calc. A1 for awrt 0.785														
MR	Allow MR of 50 (e.g. 30) provided clearly attempting $P(T \leq 2)$ and score M1A0														

Q3.

	Scheme	Marks	AO
(a)	{Let X = time spent, $P(X > 15) = \}$ 0.105649... awrt 0.106	B1 (1)	1.1b
(b)	$H_0: \mu = 10$ $H_1: \mu > 10$ $\bar{X} \sim N\left(10, \left(\frac{4}{\sqrt{20}}\right)^2\right)$; $P(\bar{X} > 11.5) = 0.046766...$ [Condone 0.9532...] [This is significant ($< 5\%$) so] there is evidence to support the complaint	B1 M1;A1 A1 (4)	2.5 3.3;3.4 2.2b
(c)(i)	$P(T < 2) =$ 0.1956... awrt 0.196	B1 (1)	1.1b
(ii)	Require $\frac{P(0 < T < 2)}{P(T > 0)} = \frac{0.119119...}{0.923436...}$; = 0.1289955... awrt 0.129	M1 A1;A1 (3)	3.4 1.1bx2
(iii)	The current model suggests non-negligible probability of T values < 0 which is impossible	B1 (1)	3.5b
(d)	Require t such that $P(T > t T > 2) = 0.5$ or $P(T < t T > 2) = 0.5$ e.g. $\frac{P(T > t)}{P(T > 2)} = 0.5$; so $P(T > t) = 0.5 \times [1 - (c)(i)]$ or $P(T > t) = 0.5 \times 0.8043...$ [i.e. $P(T > t) = 0.40...$ implies] $\frac{t-5}{3.5} = 0.2533$ or $P(T < t) = "0.5978..."$ $t = 5.886...$ or from calculator 5.867... so awrt 5.9	M1 M1; A1ft M1 A1 (5)	3.1b 1.1b 3.4 1.1b 1.1b
(15 marks)			

	Notes
(a)	B1 for awrt 0.106 (from calculator) [Allow 10.6%]
(b)	B1 for both hypotheses correct in terms of μ M1 for selection of a correct model (sight or use of correct normal- may not have label $\bar{X}$) 1 st A1 for use of this model to get probability allow 0.046~0.047 [Condone awrt 0.953] ALT OR test statistic $z = 1.677...$ (awrt 1.68) and cv of 1.64 (or better) or CR $\bar{X} > 11.47...$ 2 nd A1 (dep on 1 st A1 or at least $P(\bar{X} > 11.5) < 0.05$ (o.e.)) for a correct conclusion in context -must mention complaint/claim or time/mins is > 10 SC (M0 for $\bar{X} \sim N(11.5, ...)$ for correct probability and conclusion (score M0A0A1 on open)
(c)(i)	B1 for awrt 0.196 (from calculator) [Allow 19.6%]
(ii)	M1 for a correct probability ratio expression (may be implied by 1 st A1 scored) 1 st A1 for a correct ratio of probabilities (both correct or truncated to 2 dp) 2 nd A1 for awrt 0.129
(iii)	B1 for a suitable explanation of why model is not suitable based on negative T values Must say that a significant proportion of values < 0 (o.e.) e.g. $P(T > 0)$ should be closer to 1 or Difference between $P(T < 2 T > 0)$ and $P(T < 2)$ is too big (o.e.)
(d)	1 st M1 for a correct conditional probability statement to start the problem or $0.5 \times P(T > 2)$ 2 nd M1 for correct ratio of probability expressions [Must have $P(T > t)$ or $P(2 < T < t)$] 1 st A1ft for a correct equation for $P(T > t)$ (o.e.) ft their answer to part (c)[May be in a diagram] 3 rd M1 for attempt to find t (standardising and sight of 0.2533) or prepare to use calc (ft) Arriving at $P(T < \text{median}) = 1 - 0.5 \times "their 0.8043"$ will score 1 st 4 marks 2 nd A1 for awrt 5.9 Sight of awrt 5.9 and at least one M mark scores 5/5 [Answer only send to review]

Part	Working or answer an examiner might expect to see	Mark	Notes
(a)	$\frac{24.63 - 25}{\sigma} = -1.0364$	M1	This mark is given for standardising as part of a method to find σ
	$\sigma = 0.357$	A1	This mark is given for a correct value of σ
	$P(D > K) = 0.4$ or $P(D < K) = 0.6$	B1	This mark is given for
	$\frac{k - 25}{\sigma} = \frac{k - 25}{0.357} = 0.2533$	M1	This mark is given for using a normal model to find the probability
	$k = 25.09$	A1	This mark is given for a correct value for k
(b)	$Y \sim B(200, 0.45)$ so $W \sim N(90, 49.5)$	B1	This mark is given for setting up the normal distribution approximation of the binomial
	$P(Y < 100) \approx P(W < 99.5) = P\left(Z < \frac{99.5 - 90}{\sqrt{49.5}}\right)$	M1	This mark is given for using the normal model with a continuity correction
	$= 0.912$	A1	This mark is given for finding a correct value of the probability
(c)	$H_0 : \mu = 25$ $H_1 : \mu < 25$	B1	This mark is given for both hypotheses in terms of μ found correctly
	$\bar{D} \sim N\left(25, \frac{0.16^2}{20}\right)$	M1	This mark is given for a method to set up the normal distribution
	$P(\bar{D} < 24.94) = 0.0468$	A1	This mark is given for using the model to find a correct p -value
	$p = 0.0468 < 0.05$, so reject H_0	M1	This mark is given for a correct comparison and non-contextual conclusion
	There is sufficient evidence to support Hannah's belief	A1	This mark is given for a correct conclusion in context stated
(Total 13 marks)			

Q5.

Question	Scheme		Marks	AOs
(a)(i)	$X \sim B(15, 0.48)$		M1	3.3
	$P(X = 3) = 0.019668\dots$	awrt 0.0197	A1	3.4
(ii)	$[P(X \geq 5) = 1 - P(X \leq 4)] = 0.92013\dots$		awrt 0.920	A1
			(3)	
(b)	Y is the number of hits	M is the number of misses		
	$Y \sim N(120, 62.4)$	$M \sim N(130, 62.4)$	B1	3.3
	$P(X > 110) \approx P(Y > 110.5)$	$P(X > 110) \approx P(M < 139.5)$		
	$\left[=P\left(Z > \frac{110.5 - "120"}{\sqrt{"62.4"}}\right) \right]$	$\left[=P\left(Z < \frac{139.5 - "130"}{\sqrt{"62.4"}}\right) \right]$	M1	3.4
	$= 0.88544\dots$		A1	1.1b
			(3)	
(6 marks)				

Notes:				
(a)	M1	Writing or using the binomial distribution in (i) or (ii) Allow for sight of $B(15, 0.48)$ or in words: <u>binomial</u> with $n = 15$ and $p = 0.48$ may be implied in (i) or (ii) by one correct answer to 3sf or sight of $P(X \leq 4) = 0.07986\dots$ i.e. awrt 0.0799. Allow for ${}^{15}C_3 \times 0.48^3 \times 0.52^{12}$ as this is "correct use" Condone $B(0.48, 15)$		
(i)	A1	awrt 0.0197		
(ii)	A1	awrt 0.920 (Allow 0.92)		
(b)	B1	Setting up a correct Normal model. Allow sight of $N(120, 62.4)$ or $N(130, 62.4)$ or $N\left(120, \frac{312}{5}\right)$ or $N\left(130, \frac{312}{5}\right)$ or may be awarded if used correctly in standardisation or in words: <u>Normal</u> with <u>mean</u> = 120/130 and <u>variance</u> = 62.4 or sd = $\sqrt{62.4}$ condone $N(120, \sqrt{62.4})$ or $N(130, \sqrt{62.4})$ or sd = 62.4 Look out for $\sigma = \frac{\sqrt{1560}}{5}$ or $\frac{2\sqrt{390}}{5}$ or awrt 7.90 (condone 7.9) This may be implied by sight of 0.897 or 0.8854...		
	M1	Sight of the continuity correction with a normal distribution 110.5 or 111.5 or 109.5 139.5 or 140.5 or 138.5 NB we will also allow 129.5 or 130.5 or 128.5 NB we will also allow 120.5 or 119.5 or 121.5 Continuity correction may be seen in standardisation NB No continuity correction(CC) gives awrt 0.897 which is M0 unless CC seen		
	A1	awrt 0.8854 or awrt 0.885 dependent on sight of >110.5 or <129.5 or <139.5 or >120.5 Allow $\leq$ or $\geq$ instead of $<$ or $>$ NB 0.885548... from $B(250, 0.48)$ scores M0A0		